

6 (a) Define magnetic flux density.

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..... [2]

(b) Figure 6.1 shows a beam of charged particles in a device known as a mass spectrometer.

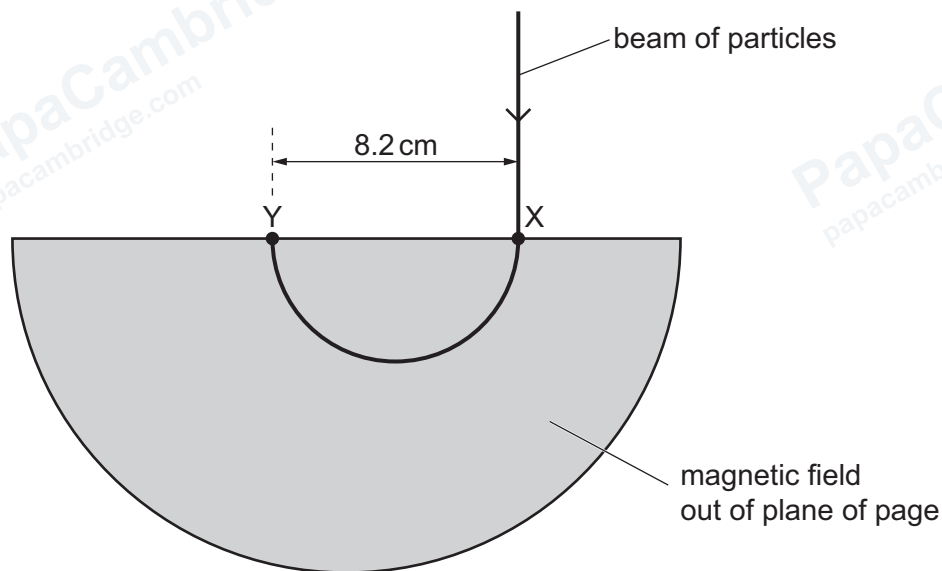


Figure 6.1

The shaded region contains a uniform magnetic field that is out of the plane of the page.

The particles enter the region of the magnetic field at point X and strike a detector at point Y.

(i) Explain whether the sign of the charge on the particles is positive or negative.

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..... [2]

- (ii) Explain why the path followed by the beam of particles inside the magnetic field is circular.

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 [2]

- (c) Each of the particles in **6(b)** has a mass of 2.0×10^{-26} kg and a charge of magnitude 3.2×10^{-19} C. All of the particles have a speed of 7.3×10^4 m s⁻¹.

The distance XY is 8.2 cm.

- (i) Show that the centripetal acceleration of one of the particles is 1.3×10^{11} m s⁻².

[2]

- (ii) Use the information in **6(c)(i)** to determine the flux density B of the uniform magnetic field. Give a unit with your answer.

$B =$ unit [2]

- (d) The particles in **6(b)** are replaced with particles that have half of the charge and a quarter of the mass of the particles in **6(b)**. The sign of the charge and the speed of the particles are both unchanged.

On Figure 6.1, draw the path of this beam of particles. [2]

[Total: 12]